

# genaut — frontier report

source CSV : genaut/logs/default.csv  
corpus rows: 929

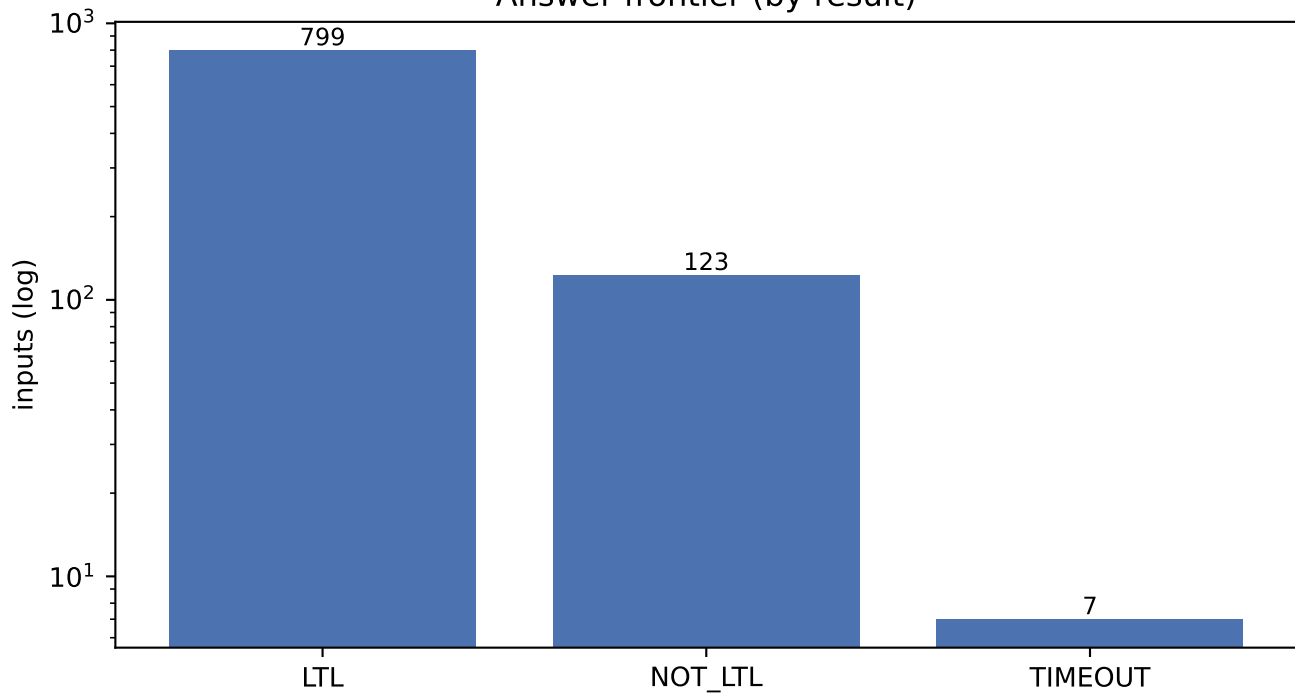
sections:

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- Routes to `true` – which technique built the universal 1
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## Ventilation — LTL / not-LTL / ambiguous

```
corpus inputs: 929
  LTL-definable (formula built) :   799  ( 86.0%)
  not-LTL (decided)             :   123  ( 13.2%)
  ambiguous (timeout, unknown)  :     7  (  0.8%)
decided (LTL or not-LTL): 922  ( 99.2%)
```

Answer frontier (by result)



## Formula collapse — structural (md5) concentration

LTL answers: 799 (53 with a gated/unprintable string)

distinct structures (md5): 121

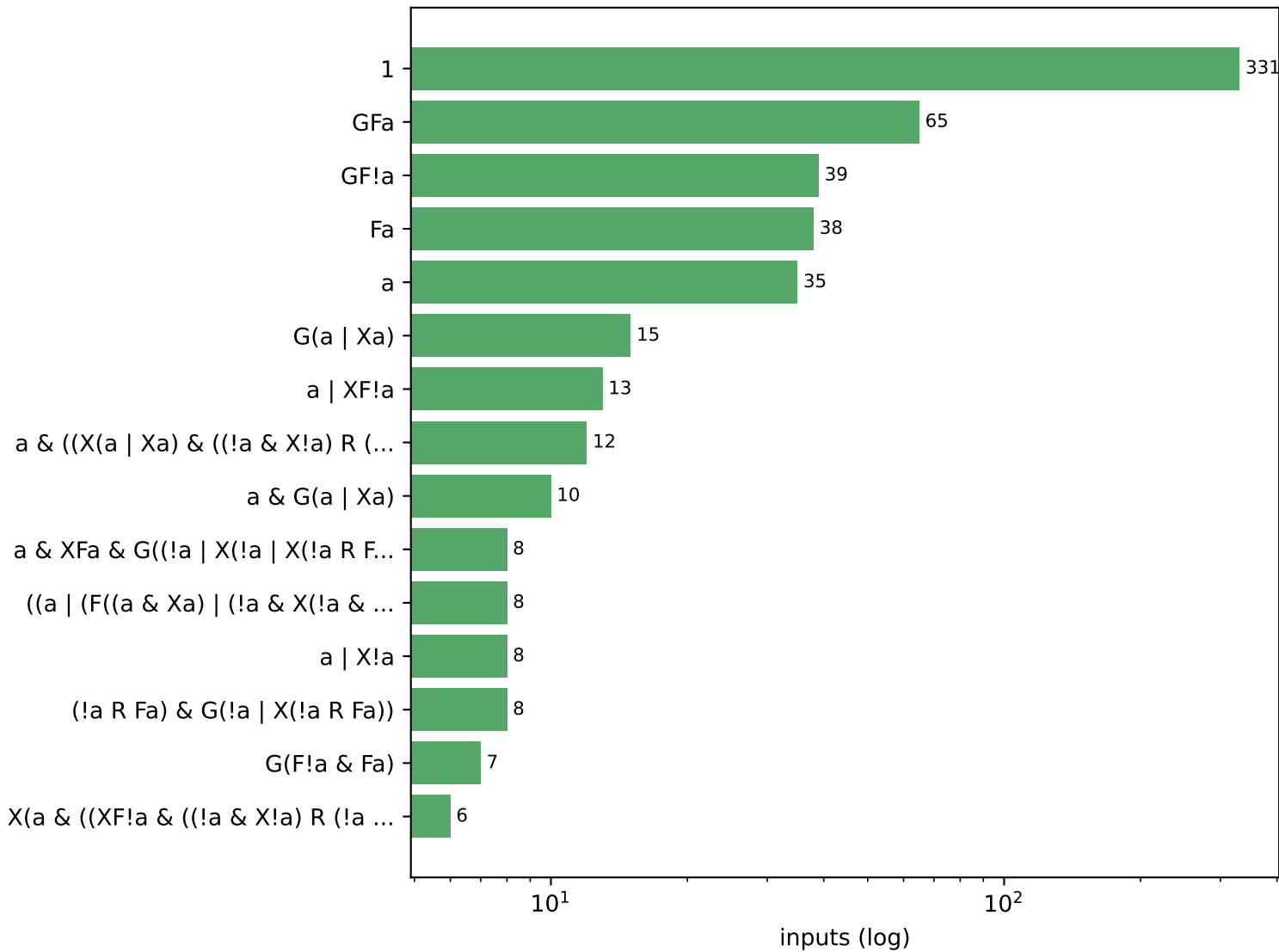
80% of answers covered by the top 22 structure(s)

90% of answers covered by the top 50 structure(s)

top 15 structures:

|     |          |                                                   |
|-----|----------|---------------------------------------------------|
| 331 | ( 41.4%) | 1                                                 |
| 65  | ( 8.1%)  | GFa                                               |
| 39  | ( 4.9%)  | GF!a                                              |
| 38  | ( 4.8%)  | Fa                                                |
| 35  | ( 4.4%)  | a                                                 |
| 15  | ( 1.9%)  | G(a   Xa)                                         |
| 13  | ( 1.6%)  | a   XF!a                                          |
| 12  | ( 1.5%)  | a & ((X(a   Xa) & ((!a & X!a) R ((!a   X(!a   ... |
| 10  | ( 1.3%)  | a & G(a   Xa)                                     |
| 8   | ( 1.0%)  | a & XFa & G(!a   X(!a   X(!a R Fa))) & (a   ...   |
| 8   | ( 1.0%)  | ((a   (F((a & Xa)   (!a & X(!a & XFa))) & (X!...  |
| 8   | ( 1.0%)  | a   X!a                                           |
| 8   | ( 1.0%)  | (!a R Fa) & G(!a   X(!a R Fa))                    |
| 7   | ( 0.9%)  | G(F!a & Fa)                                       |
| 6   | ( 0.8%)  | X(a & ((XF!a & ((!a & X!a) R (!a   X(!a R (a ...  |

Top 15 reconstructed structures



## Size frontier — flattened formula size

LTL answers with a size: 799

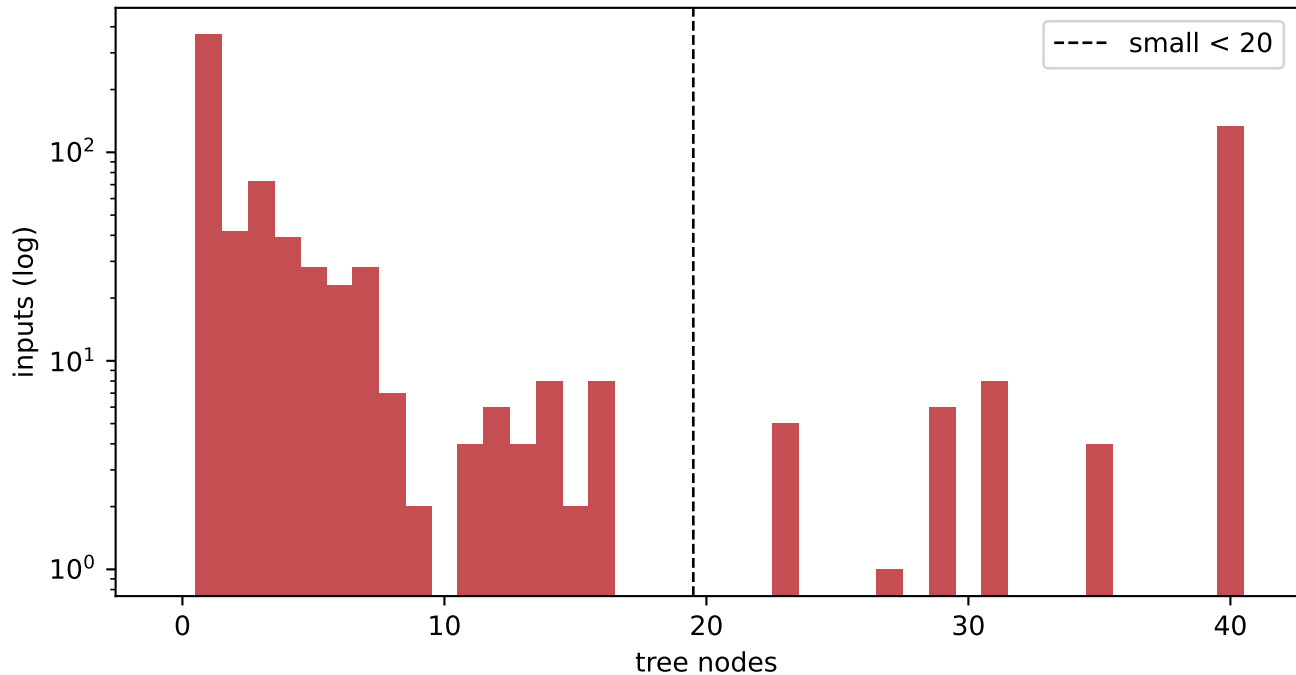
trivial (tree≤1, i.e. true/false): 367 ( 45.9%)

small (tree<20): 641 ( 80.2%)

larger (tree≥20): 158 ( 19.8%)

tree nodes: median 2 mean 132060661287776243759494201344.0 max 96278864367642945077320133640192

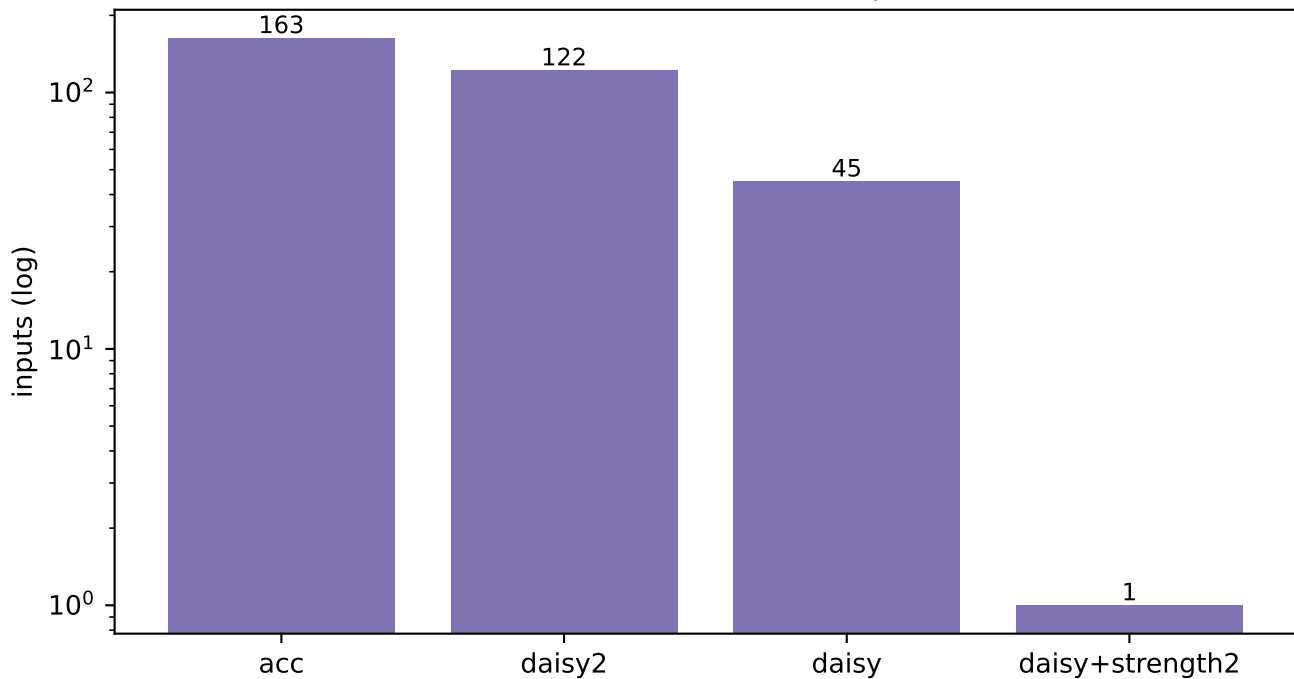
Flattened size (tree nodes); clipped at 40 (134 larger pooled at right)



## Routes to `true` — which technique built the universal 1

```
universal answers (formula == '1'): 331
  163 ( 49.2%) acc
  122 ( 36.9%) daisy2
   45 ( 13.6%) daisy
    1 (  0.3%) daisy+strength2
build_s over true rows: total 193.5s  mean 0.585s  max 1.261s
```

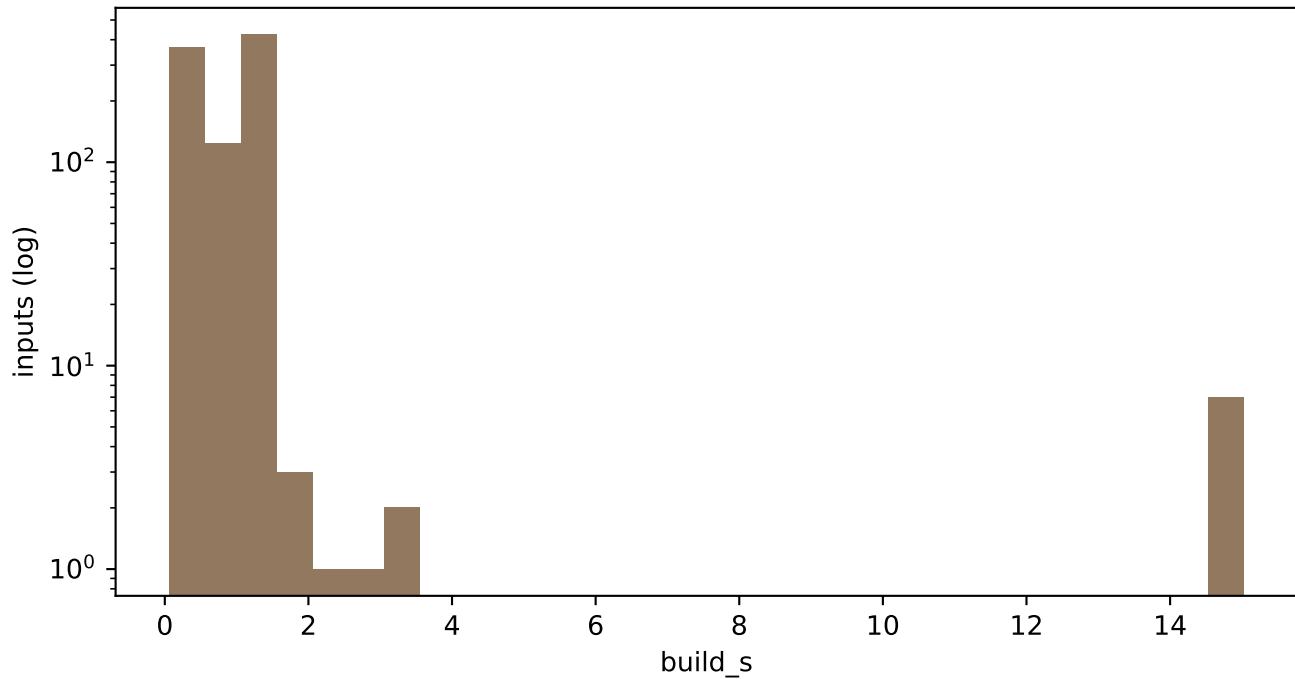
Routes to `true` (technique)



## Build time & timeouts

```
total build: 704.2s    mean 0.758s    max 15.022s
  LTL           n= 799  mean 0.657s    max 3.222s
  NOT_LTL      n= 123  mean 0.603s    max 0.665s
timeouts: 7
  aut_09412.hoa
  aut_11460.hoa
  aut_18499.hoa
  aut_18563.hoa
  aut_33916.hoa
  aut_33986.hoa
  aut_34380.hoa
```

Build-time distribution



## Verification — spot-oracle validation

```
validation over 799 LTL answers:  
  TRUE          745 ( 93.2%)  
  SIZE          53 (  6.6%)  
  TIMEOUT        1 (  0.1%)  
NOT-equivalent (FALSE): 0    CLEAN
```